Wang 2019 GBM — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

7/10 samples (≥237 cells; mult-gap 396→143) on one connected component; sample iLISI 2.60/7 across θ=5–20. Cell-cycle & metaprograms scattered — no per-sample archipelago.

Gate 2 — module geography → NO-GO

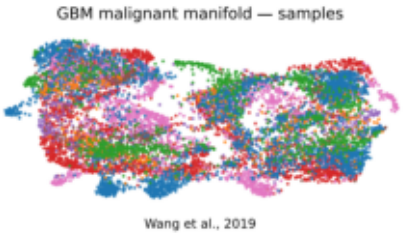
RAW module scores are ~85% noise (spatial-structure ratio 0.12–0.15 vs 0.23–0.39 in the Neftel GBM GO ref) and near-flat across all 3 lobes. Under a NEUTRAL control all three modules co-vary POSITIVELY (Gast-pSom +0.62, Crest-Gast +0.47, Crest-pSom +0.37) — no Crest opposition. The union-pool pSom↔Crest −0.61 was a control-pool artifact amplified by smoothing. No hub/arms.

Gate 3 — hub cross-repression → NA

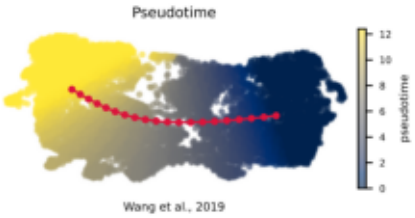
No embryonic geometry present → no trajectory forced, no bifurcation to test.

STEP FIGURES

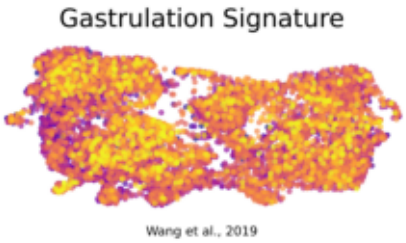
Manifold — samples



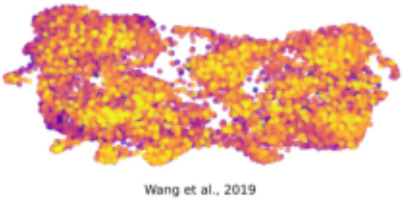
Pseudotime



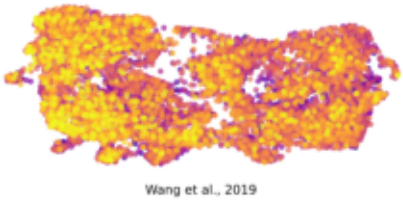
Gastrulation



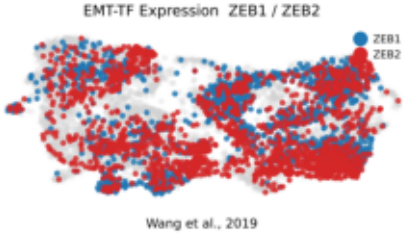
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Glioblastoma (Wang et al., 2019), 10x — 10 samples; 9,916 malignant raw → 9,650 after ≥237-cell threshold (7/10 samples) → 9,634 after island removal (16 cells, 1 detached speck).

Verdict.
✗ Pattern does NOT reproduce. The malignant manifold integrates and is continuous (Gate 1 GO), but the embryonic module geography is absent: raw Gast/pSom/Crest scores are noise-dominated and near-uniform, with no hub, no arms, and no Crest opposition once the union-pool scoring artifact is removed. Honest Gate 2 NO-GO — no trajectory forced.

Parameter choices

Min cells/sample	≥237 (mult-gap 396→143; 7/10)	**Integration**	Harmony θ=10; PCA20; spread=1.3
Embedding	native X_umap (no re-embedding)	**Gate 2 test**	raw scores + neutral-control null
Scope	whole manifold (no embryonic lobe)	**EMT-TFs**	ZEB1, ZEB2 (detectable; no order)